UNDERSTANDING AND MAINTENANCE OF NETWORKS

DAY 5: NETWORK TESTING

Setting Up and Testing a Basic Network

1. Checking Network Connectivity Using Ping

First, I connected my laptop and my friend's mobile phone to the same Wi-Fi network. I checked the IP address of both devices. Then I opened Command Prompt on my laptop and used the ping command.

Command: ping 192.168.1.5

Here, 192.168.1.5 is only an example IP address. The ping command sends small data packets to another device. If the device replies, communication is working. If it does not reply, we may see a message such as 'Request timed out'.

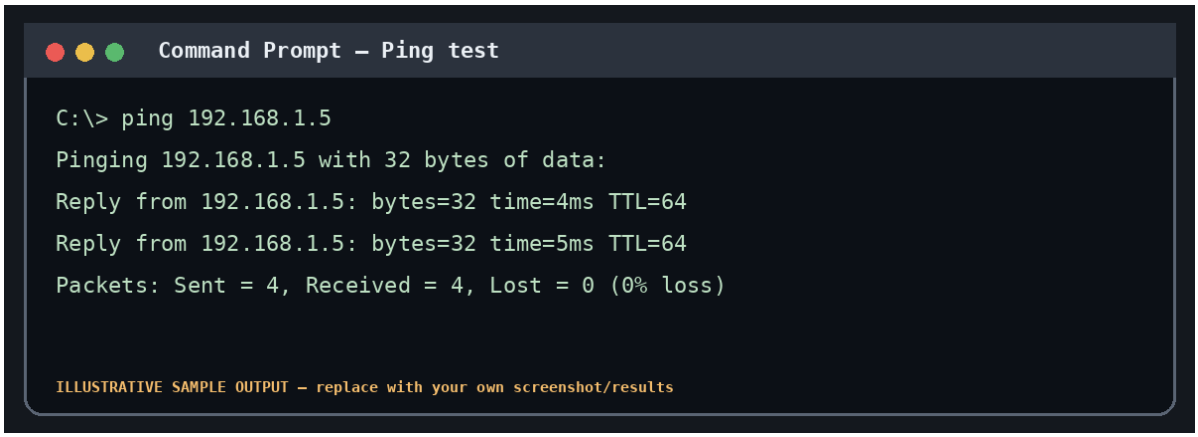


Figure 1: Sample ping command output (illustration).

2. Testing the Route Using Traceroute

I opened Command Prompt and entered the following command:

Command: tracert www.youtube.com

This command shows the path taken by data to reach the destination. Each step is called a hop. The number of hops can change depending on the internet connection and network route.

Hops observed: _____ (write the number shown on your own computer)

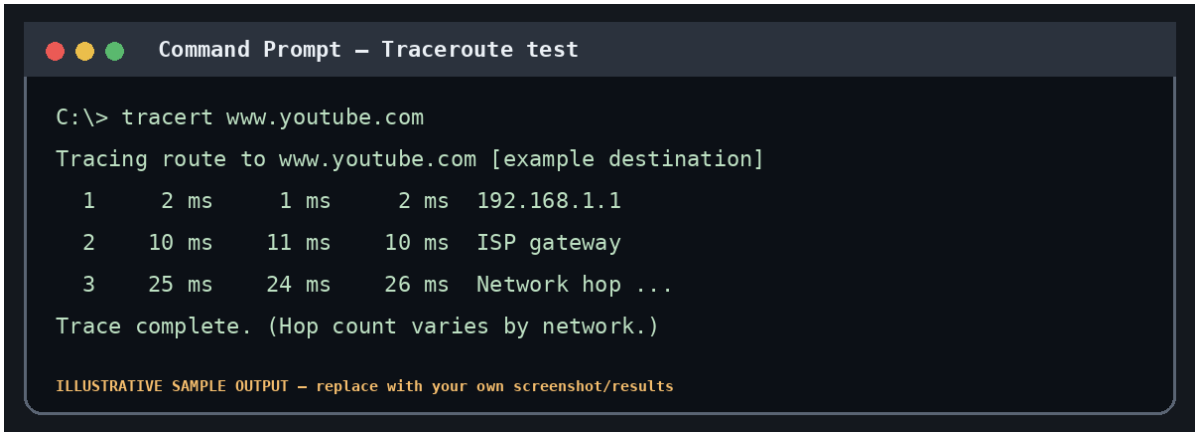


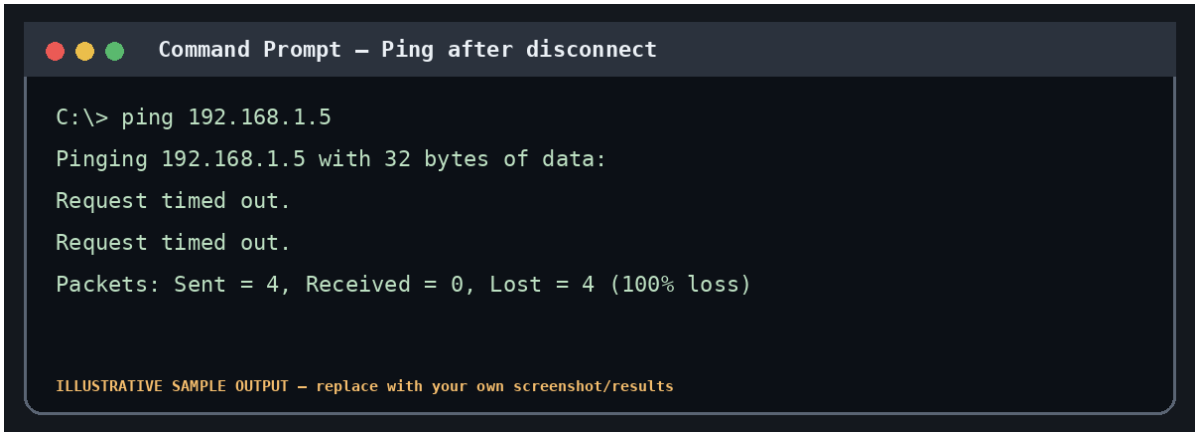
Figure 2: Sample tracert output (illustration; hop count is not a real measurement).

3. Testing After Disconnecting a Device

I disconnected my friend's phone from the Wi-Fi network. Then I tried to ping the same IP address again from my laptop.

Command: ping 192.168.1.5

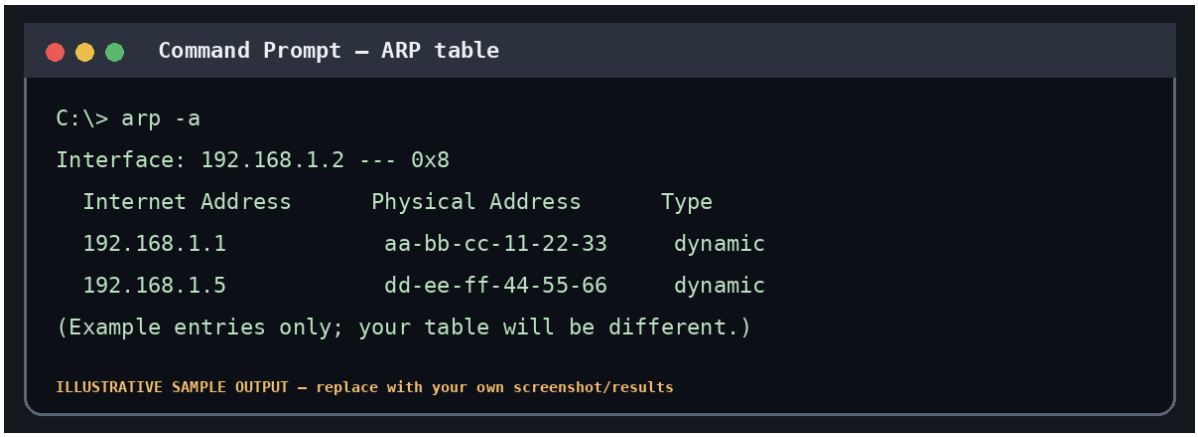
If the phone is offline and does not respond, the laptop may show 'Request timed out'. This means that no reply was received. Sometimes the phone's firewall or Wi-Fi settings can also block ping.



4. List of Devices Connected to Wi-Fi

I used the arp -a command in Command Prompt to view IP and MAC address entries known to my laptop.

Command: arp -a



No.	Device Name	IP Address	Owner
1	Wi-Fi Router	192.168.1.1	Home
2	Laptop	192.168.1.2	Me
3	Mobile Phone	192.168.1.5	Friend
4	Smart TV	192.168.1.6	Family

5. Final Report: Setting Up and Testing a Basic Network

In this practical, I learned how to connect devices to the same Wi-Fi network and check their connection using different network commands.

1. First, I connected my laptop and my friend's mobile phone to the same Wi-Fi network.
2. I checked the IP addresses of both devices.
3. I used the ping command to test whether the devices could communicate.
4. Next, I used tracert www.youtube.com to see the route taken by data and counted the hops shown.
5. I disconnected my friend's phone from Wi-Fi and tried pinging it again.
6. Finally, I used arp -a to view the IP and MAC address entries stored on my laptop.

I observed that ping is useful for checking connectivity, tracert shows the route taken by data, and arp -a displays IP-to-MAC address information known to the computer. When a device is offline, it may not reply to ping requests.